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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCBOT202 -
THALLOPHYTES AND ARCHEGONIATES
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes. Maximum Marks: 50

Part A

Answer any 10 questions (Each question carries 1 marks)

1. Name the pigment found in Sargassum.
Fucoxanthin

2. What is the specialized spore bearing structure in members of pterophyta?
sorus

3. Name the spore bearing structure in Pteris
Coenosorus-a type of sorus.

4. Which pigment gives Cladophora its green colour?
Chlorophyll a and b

5. Distinguish between the beneficial and harmful economic importance of fungi
Beneficial: Fungi are useful in food, medicine, and decomposition
Harmful: Some fungi cause plant, animal, and human diseases, food spoilage, etc

6. What is the shape of chloroplast in Volvox?
Cup shape

7. What is the role of a fungal partner in Lichen?
Provides structure, protection, and absorption.

8. Which bryophyte is commercially important for peat production?
Sphagnum or Peat Moss

9. Write one example of edible algae.
Spirulina, Chlorella (any one edible alga)

10. Give the name of a lichen which is used in beverage production.
Lethariella cladonioides, Thamnolia vermicularis, Iceland moss etc. (Any one)

11. Name a heterotrophic thallophyte.
Fungi

12. Name the asexual spores produced by fungi.
Conidiospores, Sporangiospores, Zoospores, Chlamydospores (Any Two)

Part B

Any 6 questions (Each question carries 5 marks)

13. Describe the distinguishing features of Rhodophyceae according to Fritsch

Red in colour due to phycoerythrin pigment. Storage food is floridean starch. Non- motile cells are found during reproduction. Commonly found in sea water. Sexual reproduction is oogamous type. Chlorophyll d is present in place of chlorophyll b. Examples: Polysiphonia and Batrachospermum.

14. Explain the importance of Archegoniates in plant evolution.

- Evolutionary link in the transition of plants from aquatic to terrestrial environments, showing adaptations such as a protected embryo and specialized reproductive organs.

- The presence of the archegonium, a multicellular female sex organ enclosing the egg- protecting gametes and the developing zygote from desiccation.
- Retention of the fertilized egg (zygote) within the archegonium, where it developed into a multicellular embryo, a precursor to the seed habit seen in higher plants.
- Progressive development of vascular tissues and body differentiation
- Evolutionary bridge between lower cryptogams and seed-bearing plants
- Shift in Dominant Life Cycle

15. Give an account of three subclasses of class Bryopsida as per Rothmaler's classification.

- Sphagnidae (Peat Mosses)-Primitive group including peat mosses.Possess branched gametophytes with clusters of branches at nodes. Sporophyte lacks peristome; capsule opens by an explosive lid mechanism. Example: Sphagnum.
- Andreaeidae (Granite Mosses)-Small group found mostly in cold and alpine regions.Capsules open by longitudinal slits instead of an operculum (lantern mosses). Spores are released gradually; peristome is absent. Example: Andreaea.
- Bryidae (True Mosses)-The largest and most advanced subclass. Possess a well-differentiated stem and leafy gametophyte. Sporophyte has a complex peristome with two rings of teeth (diplolepidous). Example: Funaria, Polytrichum

16. Explain the morphology of Polysiphonia.

Marine, filamentous, branched thallus that appears as delicate, feathery tufts. Reddish-brown to dark red in color and grows attached to rocks, other alga by holdfast. The plant is differentiated into a holdfast (attachment structure), main axis, and lateral branches. Polysiphonous structure – Central axial cell surrounded by pericentral cells. Regular, alternate or spiral branching; presence of trichoblasts. Cells contain Pit connections, discoid chloroplasts, cellulose wall. Lack pyrenoids. Reproductive structures (spermatangia, carpogonia, cystocarps, and tetrasporangia) develop on special branches. (5 points with description)

17. Pteridophytes are both ecologically vital and economically useful.' Evaluate this statement with examples.

Ecological Importance- Soil Formation and stabilization, Indicators of Environmental Conditions, Nitrogen Fixation

Economic Importance-Agricultural Applications-Green Manure and Biofertilizer, Pest Control. Food and Nutrition. livestock feed and fish feed. Medicinal Uses.Ornamental and Horticultural Value. Industrial Applications. Phytoremediation (Any 5 ecological and economical significance with suitable examples)

18. Explain the vegetative features of gymnosperms with suitable examples.

The main plant body is the diploid sporophyte, which is differentiated into true roots, stem, and leaves.

- well-developed tap root system. Roots often form mycorrhizal associations (e.g., *Pinus*) or coralloid roots housing cyanobacteria for nitrogen fixation (*Cycas*).
- Stems are erect, woody, and branched (*Pinus*) or unbranched (*Cycas*). They show secondary growth due to vascular cambium, forming annual rings like in dicots. Branched or unbranched. Leaf Scars: due to the shedding of old leaves.
- Vascular Tissue: The wood (secondary xylem) generally consists of tracheids but lacks vessels (except in Gnetophytes like *Gnetum*). The secondary wood is classified as: Manoxylic Wood (e.g., *Cycas*) and Pycnoxylic Wood (e.g., *Pinus*).
- Dimorphism in leaves: two types of leaves. Scale Leaves and Foliage Leavess

- Xerophytic Adaptations of leaves: Needle or Scale-like Shape,Thick Cuticle, Sunken Stomata, Transfusion Tissue (Any 5 features with explanation and examples)

19. Distinguish between vegetative cells and reproductive cells in Oedogonium.

	Vegetative Cells	Reproductive Cells
Shape	Cylindrical, uniform diameter	Zoosporangia: slightly swollen; Oogonia: distinctly swollen; Antheridia: smaller, box-like
Size	Normal, elongated	Oogonia much larger; Antheridia smaller
Function	Photosynthesis, growth	Reproduction - spore/gamete production
Contents	Single reticulate chloroplast, nucleus	Zoospores with multiple flagella; egg or sperms
	Cap cells present	Oogonia develop pores; zoosporangia rupture

20. Write any five distinguishing characters of Puccinia.

Obligate Parasitism. Polymorphism - producing up to five distinct types of spores- (Pycniospores (Spermatia), Aeciospores, Uredospores, Teliospores, Basidiospores. Heteroecious life cycle-require two different host plants. Dikaryotic Phase Dominance: specifically the Uredial stage on the primary host, responsible for the rapid, repeating spread of the disease. The presence of the characteristic Teliospore (or Teleutospore) - two-celled, thick-walled, stalked spore where karyogamy (nuclear fusion) occurs, and it functions primarily as a resting or overwintering spore. Rust fungi- Major agricultural pathogen requiring disease management strategies (5 points with explanation)

Part C

Answer any 1 questions (Each question carries 10 marks)

21. Analyze the significance of antheridia and archegonia in Riccia

- Sites of Gamete Formation
- Protection of Gametes-Both organs are multicellular and jacketed, providing protection to gametes from desiccation and mechanical injury in terrestrial habitat.
- Facilitation of Fertilization
- Site of Zygote Formation- Fertilization occurs within the archegonium
- Initiation of Sporophytic Generation-The zygote develops into the sporophyte while still attached to the gametophyte.This marks the transition from gametophytic to sporophytic phase, ensuring alternation of generations.(**Liberal valuation)**

22. Elaborate on the life cycle of Xylaria.

Xylaria is a saprophytic ascomycete (club fungus). The key structure is the stroma, an erect, dark, woody tissue that bears fructifications.

1. Mycelial Stage (Vegetative Phase)-Haploid mycelium develops from ascospores. It grows saprophytically on decaying wood or plant material.

2. Plasmogamy (Sexual Reproduction)-Two compatible hyphae fuse (plasmogamy), forming dikaryotic mycelium.
3. Formation of Stromata - Dikaryotic mycelium produces stromata (club-shaped, black, woody structures). These are the characteristic fruiting bodies visible on wood.
4. The mature stroma develops embedded, flask-shaped perithecia (sexual fruiting bodies). Each perithecium has an ostiole (pore) for spore release.
4. Spore Formation-Within the perithecia, asci develop. In each ascus (within perithecium), the nuclei fuse (karyogamy) → diploid nucleus. Meiosis follows → four haploid nuclei → mitosis → 8 haploid ascospores.
5. Germination-Ascospores are forcibly ejected and germinate to form new haploid mycelia, restarting the cycle.